ABSTRACT

HoneyChain: A Blockchain-Based System for Honey Traceability and Smart Beekeeping Management

The honey industry faces several challenges, including product adulteration, lack of supply-chain transparency, inaccurate origin information, and difficulty in verifying the authenticity and quality of honey. **HoneyChain** is a fully software-based blockchain system designed to provide secure, transparent, and traceable honey management throughout the supply chain, from beekeepers to consumers.

The system allows beekeepers and authorized stakeholders to digitally record important information such as **beekeeper details, apiary location, hive information, harvesting date, batch number, processing details, quality test results, and distribution records**. Blockchain technology provides tamper-resistant storage of critical records, making unauthorized modification of registered information difficult. Each honey batch is assigned a **unique QR code**, enabling consumers and authorized users to access its verified traceability information through a web application.

Smart contracts can manage key operations such as **batch registration, ownership transfer, verification, and supply-chain updates**. Role-based access control ensures that beekeepers, processors, distributors, administrators, and consumers can access only the information relevant to them. By combining blockchain, smart contracts, QR-based traceability, and a web-based management platform, HoneyChain aims to improve **transparency, authenticity verification, accountability, and efficient management of the honey supply chain** without requiring any dedicated hardware.